
The Muddy Children at a Muster Point

Common Knowledge from the Public Announcement
of a Fact Every Robot Already Knows

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Epistemic Robotics

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At the end of a shift N robots meet at a muster point. Each carries a status lamp, lit when its calibration has drifted, which every other robot can see and it cannot. A supervisor says over the public address that at least one of them is faulty; then a bell rings, and at each ring every robot that knows it is faulty leaves for the calibration bay. This is the muddy children puzzle, stated as an epistemic planning task whose goal is that every robot knows whether it is faulty.

With $k \geq 2$ lamps lit the announcement tells no robot anything it did not know: each sees a lit lamp. What it changes is the depth of the group's knowledge of that fact, from E^{k-1} to common knowledge. We prove that without the announcement no robot ever learns its own lamp, so that every bell leaves the model as it was and no policy exists at any depth; that with it the k faulty robots know after $k - 1$ silent bells and leave at the k -th, at which the others learn that they are not faulty; and that a policy of depth N exists and none of smaller depth. Aletheia agrees for two to ten robots: a policy of depth N with N leaves, found in $3N(N + 1)/2$ expansions, and no policy without the public address, which its relaxation proves. The report is the counterpart of the coordinated attack, where common knowledge was a precondition no private message could supply; here a public event supplies it, and that is what makes the bells informative. Four robots were run on both floors in Gazebo through ePlanSys: with the announcement the three faulty robots left at the third bell and the fourth went back to work; without it four bells passed and nobody moved.

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1 Scope

The companion report on the coordinated attack [6] states common knowledge as the precondition of a joint action, proves that a lossy radio cannot supply it, and shows a public signal supplying it in one update. This report takes a problem in which common knowledge is not a precondition stated in the domain, and is still what an epistemic policy turns on: the muddy children [4, 5, 2], in the form of robots at a muster point with status lamps they cannot see on themselves.

Four claims are advanced.

1. Before the announcement, at a world with k lamps lit, “some lamp is lit” holds at depth exactly E^{k-1} (Lemma 3); every robot knows it as soon as $k \geq 2$, and it is common knowledge only after the announcement.
2. Without the announcement no robot ever learns its own lamp: every bell is the identity update, and no policy exists, of any depth (Proposition 1).
3. With it, k faulty robots know after $k - 1$ silent bells and leave at the k -th, at which the others learn that they are not faulty (Proposition 2); a policy of depth N with N leaves exists, and none of smaller depth (Proposition 3).
4. The planner agrees, for two to ten robots, on both floors, and both floors were executed with four robots in Gazebo through ePlanSys, the bell’s outcome given by the robots that know at the actual world (Section 10).

Three bounds are stated at the outset. The lamps are the fleet’s diagnostics, and the robots’ observation of each other’s lamps is a modelling assumption, checked on the floor plan as a line of sight between every pair of places on the muster ring and not perceived by a camera. The bell observes whether any robot left, not which robots did; Section 6 shows that the model needs no more. And the robots do not reason for themselves: which of them know they are faulty is read, at each bell, from the model the epistemic state maintains, evaluated at the world the lamps make actual.

2 Preliminaries

The definitions are those of the companion report: epistemic models $\mathcal{M} = (W, R, V, W_d)$, the group modalities E_G and C_G over $R_G = \bigcup_{a \in G} R_a$, and event models with observability types and their product update. We recall the two facts used below.

Definition 1 (Distance and depth). For $\mathcal{M}, w \models \varphi$, $\text{dist}_w(\varphi)$ is the length of the shortest R_G -path from w to a world where φ fails, and ∞ if there is none; $\text{depth}_w(\varphi) = \text{dist}_w(\varphi) - 1$.

Lemma 1 ([6]). If $\mathcal{M}$ is S5 and $\mathcal{M}, w \models \varphi$, then for $k \geq 1$, $\mathcal{M}, w \models E_G^k \varphi$ iff $k \leq \text{depth}_w(\varphi)$, and $\mathcal{M}, w \models C_G \varphi$ iff $\text{dist}_w(\varphi) = \infty$.

Definition 2 (Public sensing). For a formula ψ , $P(\psi)$ has events e^+ with $\text{pre}(e^+) = \psi$ and e^- with $\text{pre}(e^-) = \neg\psi$, trivial postconditions, both designated, and $Q(\tau) = \{(e^+, e^+), (e^-, e^-)\}$ for every agent.

Lemma 2 (Public sensing restricts). Let $(w, e) \in \mathcal{M} \otimes P(\psi)$. The worlds of $\mathcal{M} \otimes P(\psi)$ reachable from (w, e) , with their relations and valuations, are those of the restriction of $\mathcal{M}$ to $\{v : \mathcal{M}, v \models \text{pre}(e)\}$ reachable from w .

Proof. $(v, f) R'_a (u, g)$ iff $v R_a u$ and $f = g$, so every world reachable from (w, e) has the event e , and (u, e) exists iff $\mathcal{M}, u \models \text{pre}(e)$. The postconditions are trivial. $\square$

Both actions of the domain are of this form. The announcement is $P(\theta)$ with $\theta = \bigvee_i \text{faulty}(i)$, “some lamp is lit”, and its event e^- can occur only at the world with no lamp lit, which is never designated. The bell is $P(\lambda)$ with $\lambda = \bigvee_i K_i \text{faulty}(i)$, “some robot knows that it is faulty”: e^+ is **e-leave** and e^- is **e-stay**.

3 The domain

`tools/muster.py` writes the domain for N robots $r_1, \dots, r_N$ and two floors that differ in one line, whether the hall has a public address, pa . The atom $faulty(i)$ says that r_i 's calibration has drifted and its lamp is lit. Every robot sees every other robot's lamp and not its own, and that is common knowledge; that some lamp is lit is true, and not common knowledge. The goal is that every robot knows whether it is faulty:

$$\gamma = \bigwedge_i Kw_{r_i} faulty(i).$$

Table 1: The actions of the domain.

action	type	events	observers
<i>announce</i>	public announcement	$pa \wedge \theta; pa \wedge \neg\theta$	everyone
<i>bell</i>	public sensing	e-leave: $\bigvee_i K_i faulty(i)$; e-stay: $\bigwedge_i \neg K_i faulty(i)$	everyone

```
(:event e-leave
:parameters (?h - hall)
:precondition (exists (?i - agent) ([?i] (faulty ?i))))

(:event e-stay
:parameters (?h - hall)
:precondition (forall (?i - agent) (not ([?i] (faulty ?i)))))

(:action bell
:parameters (?h - hall)
:action-type (public-sensing (e-leave ?h) (e-stay ?h))
:observability-conditions (default Fully))
```

Listing 1: The bell, in the domain `tools/muster.py` writes.

Why the bell observes only whether anyone left. On the floor every robot sees which robots drive off. The model takes less, a public sensing of λ , and Section 6 shows that it loses nothing: after the first departure every robot already knows whether it is faulty, so who left adds no information the goal needs.

Why the hall is a parameter. `plank` grounds an action once for each assignment of its parameters, and an action with none is never grounded. Both actions take the hall, of which each problem has one, `muster`.

4 The model

Definition 3 (The muster model). $\mathcal{H}_N = (W, R, V, W_d)$ with $W = \{0, 1\}^N$, uR_iv iff $u_j = v_j$ for every $j \neq i$, $V(u) = \{faulty(i) : u_i = 1\} \cup V_{pa}$, and $W_d = W \setminus \{\mathbf{0}\}$. For $u \in W$ let $|u|$ be the number of lamps lit, and u^i the world that differs from u in the i -th lamp only.

R_i relates two worlds exactly when they differ at most in r_i 's own lamp, which is the statement that r_i sees every lamp but its own. Each R_i is an equivalence with classes $\{u, u^i\}$, and $\mathcal{H}_N$ is S5. `plank` grounds the initial model of the problem to $\mathcal{H}_N$: 2^N worlds, of which $2^N - 1$ designated. The designated worlds are the planner's uncertainty about the lamps, not any robot's; what a robot knows is what it knows at the actual world, the one whose lit lamps are the faults.

Lemma 3 (The depth before the announcement). For $u \in W_d$ with $|u| = k$, $\text{dist}_u(\theta) = k$ in $\mathcal{H}_N$: the robots have $E_G^{k-1}\theta$ at u and not $E_G^k\theta$.

Proof. θ fails only at $\mathbf{0}$. An R_G -step changes at most one lamp, so a path from u to $\mathbf{0}$ has at least k steps; turning the k lit lamps off one at a time, each step by the robot whose lamp it is, is such a path of k steps. Lemma 1 gives the rest. $\square$

With $k = 1$ the faulty robot does not know that any lamp is lit, since it sees none. With $k \geq 2$ every robot knows it, and the depth E^{k-1} falls short of common knowledge by a chain through the faulty robots: r_1 considers that only $r_2, \dots, r_k$ are lit, in which world r_2 considers that only $r_3, \dots, r_k$ are, and so on down to the world with no lamp lit.

5 Silence

Proposition 1 (Without the announcement, no robot learns its lamp). In $\mathcal{H}_N$ no robot knows whether it is faulty, at any world: $\mathcal{H}_N, u \not\models K_i \text{faulty}(i)$ and $\mathcal{H}_N, u \not\models K_i \neg \text{faulty}(i)$ for every u and i . Hence the bell's **e-leave** occurs nowhere, **e-stay** occurs everywhere, and the part of $\mathcal{H}_N \otimes \text{bell}$ reachable from any world is $\mathcal{H}_N$ again.

Proof. $uR_i u^i$, and u and u^i disagree on $\text{faulty}(i)$. So λ fails at every world, and by Lemma 2 the **e-stay** copy of $\mathcal{H}_N$ is $\mathcal{H}_N$, restricted to all of W . $\square$

Corollary 1 (No policy without the public address). On the floor without a public address no sequence of actions makes γ hold, and no policy exists, of any depth.

Proof. *announce* requires *pa*, which is false, so every execution is a sequence of bells. By Proposition 1 and induction each leaves the model $\mathcal{H}_N$, at which γ fails at every world. $\square$

Every robot can see, at a world with three lamps lit, that some lamp is lit, and the bell still teaches nothing. The reason is the chain of Lemma 3: that nobody left is what every robot expected in every world it considers possible, including the world with no lamp lit at the end of the chain, where nobody could have left.

6 The announcement and the bells

For $0 \leq j < N$ let $\mathcal{M}_j$ be $\mathcal{H}_N$ restricted to $\{u : |u| \geq j + 1\}$, and L_j be $\mathcal{H}_N$ restricted to $\{u : |u| = j\}$.

Lemma 4 (The announcement). The part of $\mathcal{H}_N \otimes \text{announce}$ reachable from a designated world is $\mathcal{M}_0$, and $\mathcal{M}_0 \models C_G \theta$.

Proof. At a designated world the event that occurs is the one with θ , and by Lemma 2 the reachable part is $\mathcal{H}_N$ restricted to the worlds with a lamp lit, that is $\mathcal{M}_0$. θ holds at all of them, so its distance is infinite. $\square$

Proposition 2 (The bells after the announcement). In $\mathcal{M}_j$, at u :

1. r_i knows that it is faulty iff $u_i = 1$ and $|u| = j + 1$, and no robot knows that it is not faulty;
2. the bell gives **e-leave** iff $|u| = j + 1$, and then the reachable part of the update is L_{j+1} , in which every robot knows whether it is faulty; otherwise it gives **e-stay**, and the reachable part is $\mathcal{M}_{j+1}$.

Proof. In $\mathcal{M}_j$ the class of u for r_i is $\{u, u^i\}$ if $|u^i| \geq j + 1$, and $\{u\}$ otherwise. If $u_i = 0$ then $|u^i| = |u| + 1$, so u^i is present, and r_i does not know its lamp. If $u_i = 1$ then $|u^i| = |u| - 1$, which is below $j + 1$ exactly when $|u| = j + 1$; then r_i 's class is $\{u\}$ and it knows that it is faulty. This is the first part. So λ holds at u iff $|u| = j + 1$, and by Lemma 2 the **e-leave** part is $\mathcal{M}_j$ restricted to $|u| = j + 1$, which is L_{j+1} , and the **e-stay** part is $\mathcal{M}_j$ restricted to $|u| \geq j + 2$, which is $\mathcal{M}_{j+1}$. In L_{j+1} two distinct worlds have the same number of lamps lit and so differ in at least two, and no relation joins them: every class is a single world, and every robot knows its lamp. $\square$

Corollary 2 (k faulty robots). At a world with k lamps lit, after the announcement the first $k - 1$ bells give **e-stay**. After the $(k - 1)$ -th the k faulty robots know that they are faulty; the k -th gives **e-leave**, and after it the others know that they are not. With $k = N$, $\mathcal{M}_{N-1}$ is the single world **1**, and every robot knows after $N - 1$ bells.

Proposition 3 (The depth of a policy). On the floor with a public address the policy that announces, and then rings the bell until it gives **e-leave** or has rung $N - 1$ times, satisfies γ at every leaf; it has depth N and N leaves. No policy of smaller depth exists.

Proof. By Corollary 2, the branch on which the j -th bell first gives **e-leave**, for $j \leq N - 1$, ends in L_j , and the branch of $N - 1$ **e-stay** ends in $\mathcal{M}_{N-1} = \{\mathbf{1}\}$; γ holds at every world of each. That is N leaves, and the longest branch has $1 + (N - 1)$ actions. For the lower bound take the designated world **1**. Bells before the announcement leave the model unchanged (Proposition 1), so the branch a policy follows at **1** must announce, and after the announcement and j bells, all of which give **e-stay** at **1** while $j \leq N - 1$, the model is $\mathcal{M}_j$. For $j < N - 1$, r_i considers $\mathbf{1}^i$, with $N - 1 \geq j + 1$ lamps lit, so it does not know its lamp at **1** and γ fails there. The branch therefore has at least $1 + (N - 1)$ actions. $\square$

With $k \geq 2$ the announcement gives no robot a fact it lacked, and Proposition 1 shows that the bells alone give nothing. What the announcement gives is the depth: it turns $E^{k-1}\theta$ into $C\theta$. The first silent bell is then informative, because after the announcement a world with one lamp lit is one in which that robot knows it is faulty and leaves, so that nobody leaving rules every such world out.

7 Planning results

Both floors were grounded by plank and solved by Aletheia with AO* at one thread, for two to ten robots. On the floor without a public address Aletheia's knowledge relaxation proves the goal unreachable before any search, which Corollary 1 confirms. On the floor with one it returns the policy of Proposition 3, of depth N with N leaves; the trace tool replays the policy for four robots and reports γ at every designated world of each of the four leaves, covering the fifteen patterns with a lamp lit.

Table 2: Both floors for N robots. Times are wall-clock for the planner process, loading included.

N	worlds	public address				none	
		depth	leaves	expansions	time	result	time
2	4	2	2	9	0.003 s	unreachable	0.003 s
3	8	3	3	18	0.003 s	unreachable	0.003 s
4	16	4	4	30	0.004 s	unreachable	0.003 s
5	32	5	5	45	0.006 s	unreachable	0.004 s
6	64	6	6	63	0.012 s	unreachable	0.004 s
7	128	7	7	84	0.031 s	unreachable	0.007 s
8	256	8	8	108	0.071 s	unreachable	0.007 s
9	512	9	9	135	0.239 s	unreachable	0.014 s
10	1 024	10	10	165	0.690 s	unreachable	0.023 s

The expansions are $3N(N + 1)/2$ for every N measured; that is an observation, not a property proved here. The time grows with the model, which doubles with each robot, and not with the search, which grows quadratically.

8 Four robots, three lamps lit

Table 3 follows the actual world with r_1 , r_2 and r_3 lit, on both floors, as the trace tool computes it from the task plank grounds.

Table 3: Four robots with r_1 , r_2 and r_3 lit. Worlds are those reachable from the actual world; a robot is marked when it knows its lamp.

floor	action	$ W $	knows its lamp	depth of θ
public address	initial	16	none	E^2
	<i>announce</i>	15	none	C
	<i>bell</i> $\rightarrow$ e-stay	11	none	C
	<i>bell</i> $\rightarrow$ e-stay	5	r_1, r_2, r_3 : faulty	C
	<i>bell</i> $\rightarrow$ e-leave	1	all; r_4 : not faulty	C
none	initial	16	none	E^2
	<i>bell</i> $\rightarrow$ e-stay , four times	16	none	E^2

The counts are those of Section 6: $\mathcal{M}_0$ has $2^4 - 1 = 15$ worlds, $\mathcal{M}_1$ the eleven with two or more lamps lit, $\mathcal{M}_2$ the five with three or four, and L_3 has four worlds, none related to another, so that one is reachable. Before the announcement the chain that blocks common knowledge runs through the three faulty robots, as Lemma 3 describes. With only r_2 lit, r_2 knows that it is faulty from the announcement alone, and the first bell gives **e-leave**.

9 The floor

The floor is the AWS RoboMaker small warehouse, the world of the robot-warehouse demonstrations, unchanged, with a muster point, a calibration bay and a public address added on its open floor (Figure 1). The floor plan is rasterised from the world’s own collision meshes at 5 cm. The muster point is the largest open circle on the floor, with 3.6 m of clearance; the robots stand on a ring of 1.2 m about its centre, facing in. The calibration bay is in the room at the north-west corner, and each robot’s station, where it starts and returns to, is at the end of the west or the middle corridor.

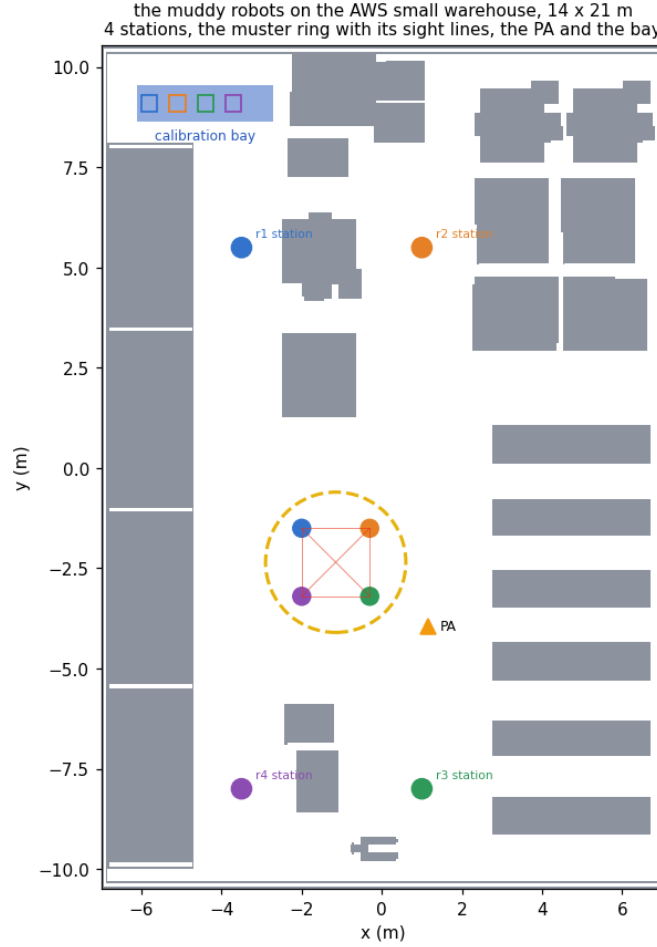


Figure 1: The floor plan with four robots: their stations, the muster ring with the sight lines between every pair of places on it, the public address, and the calibration bay with a place for each robot.

The domain makes two claims about the floor, and the floor check tests each on the plan, by casting rays over the occupied cells or flooding the free ones (Table 4). It passes for three, four and five robots.

Table 4: The claims about the floor, each checked on the plan.

claim	what depends on it
on the muster ring every robot has every other robot in line of sight	the initial model, in which each robot knows whether every other is faulty
every place a robot is sent is reachable from where it is sent from, over the plan inflated by 0.35 m	a correct policy, and the dismissal after it, are drivable

The actions are performed by one node that owns every robot’s driver. Before planning it drives the robots from their stations to the ring. The announcement lights a lamp on the public-address mast. At the bell the node waits for the model that includes the previous update, finds the robots that know they are faulty at the actual world, and refuses if the model says so of a robot whose lamp is dark, or if no designated world matches the lamps. It then sends the robots that know to the bay one at a time, nearest first, and reports *e-leave*, or reports *e-stay* if there are none. After the policy every robot is dismissed on what it knows: the faulty to the bay, the others back to their stations, and the dismissal is refused if the model’s verdict on any robot disagrees with its lamp.

10 Runs

Both floors were run with four robots and r_1 , r_2 and r_3 lit, and recorded on a virtual display with RViz and Gazebo side by side. In the tables the model column gives the worlds the robots can still consider, those reachable from the actual world, then the patterns the executor cannot rule out, the designated worlds, and the depth of “some lamp is lit” at the actual world.

Table 5: The floor with a public address, as the run reported it.

action	reported	model
assembly	every robot at its place on the ring after 16 to 25 s, each in sight of every other robot’s lamp	16 / 15, E^2
plan	4 policy items, 4 leaves, 0.1 s	
announce	“at least one of you is faulty”, over the public address	15 / 15, C
bell	no robot knows whether it is faulty, and nobody moves: e-stay	11 / 11, C
bell	nobody moves: e-stay ; r_1 , r_2 and r_3 now know that they are faulty	5 / 5, C
bell	r_1 , r_2 and r_3 know that they are faulty and leave, one at a time: e-leave ; in the bay after 34, 41 and 48 s	1 / 4, C
dismissal	r_4 knows that it is not faulty and goes back to its station, after 20 s; the three are in the bay	

Table 6: The floor without a public address, as the run reported it.

action	reported	model
assembly	every robot at its place on the ring after 17 to 24 s	16 / 15, E^2
plan	no policy, after 0.2 s; the mission rings the bell four times by hand	
bell, four times	no robot knows whether it is faulty, and nobody moves: e-stay	16 / 15, E^2
dismissal	no robot knows whether it is faulty, and every robot stays where it is	

The counts are those of Table 3, and the designated worlds after each bell are those of Proposition 2: the fifteen patterns with a lamp lit, then the eleven with two or more, the five with three or four, and after **e-leave** the four with exactly three, at each of which every robot knows its lamp. On the floor without a public address the designated worlds stayed at fifteen through four bells, as Proposition 1 says, and the mission confirmed through the epistemic state that no robot knew whether it was faulty before it reported that as the expected outcome.

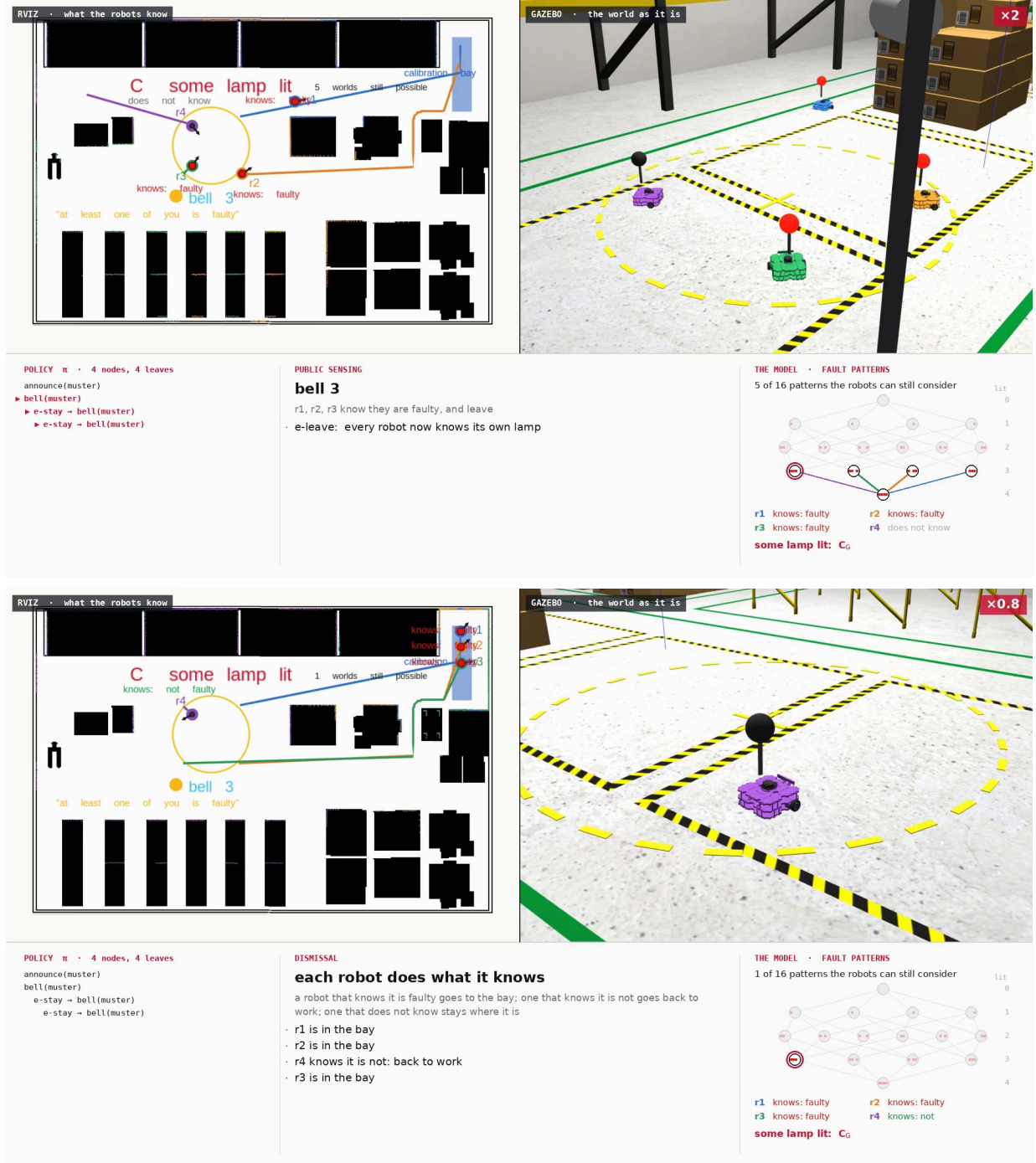


Figure 2: Two frames of the film. Above, the third bell: $r1, r2$ and $r3$ know that they are faulty and leave one at a time, $r4$ does not know, and five patterns remain, the four with three lamps lit and the one with four. Below, the dismissal: one pattern remains for the robots, $r4$ knows that it is not faulty and drives back to its station, and the three are in the bay.

11 Defects

Knowledge judged at the designated worlds. The first run took a robot to know that it was faulty when it did at every designated world, the fifteen patterns the executor could not rule out, and not at the actual one. After two silent bells $r1$ knows that it is faulty at the actual world, where $r1, r2$ and $r3$ are lit, and not at the world where $r2, r3$ and $r4$ are; under that reading no robot knew, and the third bell reported e-stay. The epistemic state applies the outcome a performer reports, and e-stay removed every world with three lamps lit, the actual world among

them. The model was left with the world in which all four are lit, the dismissal sent r_4 , whose lamp was dark, to the bay, and the mission reported the goal achieved, since in that model every robot knew whether it was faulty. The designated worlds are the planner's uncertainty about the lamps and not any robot's. The crew now evaluates knowledge at the actual world, the bell refuses when no designated world matches the lamps, and the dismissal refuses when the model and a lamp disagree.

Three robots leaving at once. When the three faulty robots left the ring together they drove into one another across it: r_1 reached the bay after 30 s, r_2 after 32 s and r_3 after 243 s. A robot leaving the ring now steps out along its radius first, and the robots leave one at a time, nearest the bay first, five seconds apart; in the next run they arrived after 31, 36 and 41 s.

A goal written as a sentence. On the floor without a public address the mission rings the bell by hand, since there is no policy, and the policy it builds for that carried its goal as an English sentence, which the behaviour tree parsed as a formula. It is now $\bigwedge_i Kw_{r_i} \text{faulty}(i)$, and the executor's report that the goal does not hold at the end is the expected outcome on that floor.

Guards added on review. The bell refuses when no model has arrived, and when no state message newer than the end of the previous action arrives within 10 s, so that it never reads a model that lacks an update. A formula check the epistemic state cannot answer fails the mission, where it had been counted as a robot not knowing; on the floor without a public address that would have read as the expected outcome.

12 The counterpart of the coordinated attack

In the coordinated attack, common knowledge of the stand is the precondition of the lift, and the question is which channel can supply it: a lossy message adds at most one level of mutual knowledge, so no number of them can, and one public signal seen from common viewpoints does. Here common knowledge is not asked for by any precondition, and the same distinction decides the outcome. The robots begin with $E^{k-1}\theta$, which a finite exchange of messages could have raised a level at a time; the bells, public events, teach nothing at that depth (Proposition 1); and one public announcement, of a fact every robot already knows, gives $C\theta$, after which each public bell removes one layer of the model.

Halpern and Moses treat both puzzles in their account of common knowledge in distributed systems [3]. The two domains place the public event differently: at the end of the coordinated attack, where it makes the lift applicable, and at the start of the muster, where it makes the bells informative.

13 Directions for the next iteration

An announcement over the radio. If the supervisor's message reached each robot over the lossy channel of the coordinated attack, it would add at most one level of mutual knowledge to θ , and common knowledge would not arise. Which fault patterns the bells would then settle, as a function of the depth the message reaches, is a question the domain can put to the planner.

A robot that cannot see one other. If a pillar hides r_j 's lamp from r_i , R_i relates worlds that differ in either lamp, the model is no longer the hypercube, and the number of bells a muster needs depends on the pattern of occlusion; the floor check would then be a statement about which pairs see each other and not a requirement that all do.

References

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Reproduction

```
python3 tools/muster.py --robots 4 --out /tmp/muddy # the domain and both floors
bash tools/validate.sh # seven checks, no simulator
python3 tools/trace.py --task /tmp/muddy_robots_validate/n4/pa/pa-n4.json --faults r1 r2 r3 --bells 4
python3 tools/check_floor.py --robots 4
```